

HOUSE PRICE PREDICTION MODEL - PROJECT REPORT

OBJECTIVE:

Build a regression model to predict house prices based on property features.

KEY FINDINGS:

1. MOST INFLUENTIAL FEATURES:

The top features affecting house prices are:

- area: 46.84%
- bathrooms: 15.15%
- airconditioning_yes: 6.27%
- parking: 5.75%
- stories: 5.71%

2. MODEL PERFORMANCE:

Our Random Forest model achieved an R^2 score of 0.6119, explaining 61.2% of price variance.

The average prediction error is \$1,021,546.

3. DATA INSIGHTS:

After cleaning, we analyzed 545 properties with 14 features.

Average house price: \$4,766,729

Price range: \$1,750,000 - \$13,300,000

4. MODEL COMPARISON:

Random Forest outperformed Linear Regression, showing non-linear relationships are important for price prediction.

RECOMMENDATION:

Real estate businesses should use this Random Forest model for accurate pricing.

Focus marketing on the top 5 features to improve sales strategies.

Analysis completed successfully!